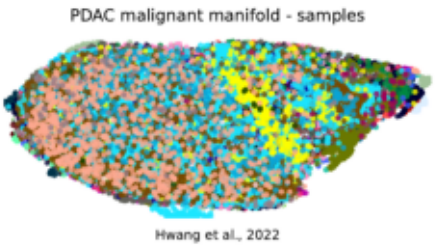
CCCA Cell-State Analysis — Hwang et al. 2022, PDAC (snRNA-seq / 10x)

GO / NO-GO GATES

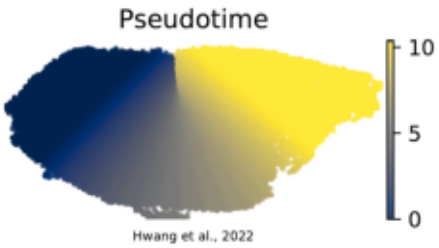
Gate 1	**GO**	samples mixed (iLISI 9.3/38 vs 1.0 raw; falls to 6.7 at theta=20) on one continuous connected manifold; density lobes are poles of one axis, not batch islands nor a cycling compartment
Gate 2	**GO**	organized single-axis geography: Gast high at root, opposes pSom/Crest spatially (Gast<->pSom r=-0.33); Crest an inverse/terminal gradient. Non-branching case (§1) -> GO
Gate 3	**NA**	no bifurcation (scFates curve, not tree) -> cross-repression scatter N/A

STEP FIGURES

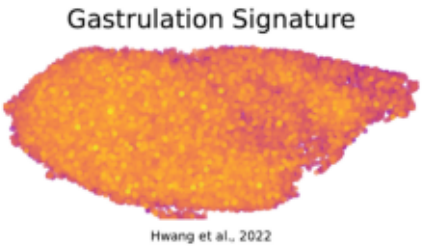
01 Samples (integration)



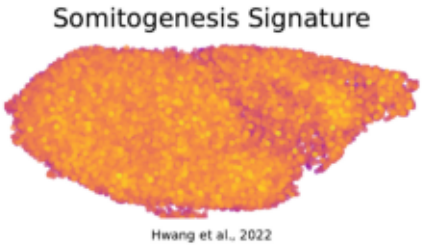
02 Pseudotime (curve)



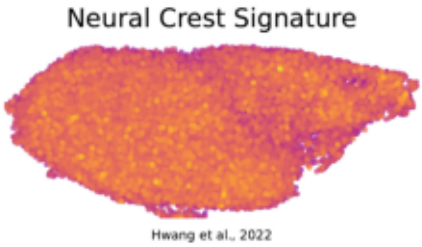
03 Gast signature



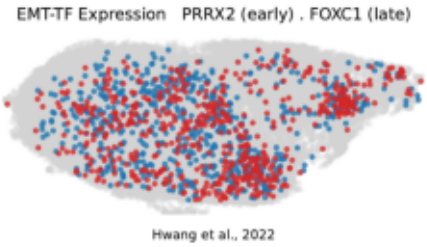
03 pSom signature



03 Crest signature



04 EMT-TF expression



DATASET

Pancreatic ductal adenocarcinoma (PDAC), single-nucleus RNA-seq / 10x (Hwang et al., 2022).
64,401 malignant cells across 38 tumours after threshold + island removal.

VERDICT

Pattern reproduces as the NON-BRANCHING case: one continuous gastrulation -> somitogenesis / neural-crest axis, rooted at the Gast/pSom-high developmental pole. No competing arms and no cross-repression bifurcation. Geometry = Linear-pSom/Crest.

KEY PARAMETER CHOICES

Malignant cells	64,538 raw -> 64,507 (thresh) -> 64,401 (islands)
Min-cell threshold	14 (gap-finding; smooth tail, only 24->9 gap); 5 tiny samples dropped
Integration	Harmony on sample, theta=10, n_pcs=20, spread=1.3
Overdispersed genes	1,664 (of 19,457 retained)
Geometry	Linear-pSom/Crest (curve: 2 tips, 0 forks, fold-free 2.72)
Root	Gast/pSom-high tip (developmental origin; cycling flat, rho=-0.03)
Module means	Gast -0.005 | pSom -0.002 | Crest +0.003 (weak, spatially organized)
Crest arm	no inversion (Crest opposes Gast, r=-0.30)
EMT-TFs shown	PRRX2 (root) . FOXC1 (terminus); ZEB1 most expressed (~16%)
Hub / r	n/a (non-branching)